

Get Vari: Product Requirements & Working Report

Section 1: Understanding Hydration Parameters (Primary & Secondary)

Research Paper-1: Towards Data-Driven Hydration Monitoring: Insights from Wearable Sensors and Advanced Machine Learning Techniques(2024)

Research Paper-2: Multispectral sensor fusion in SmartWatch for in situ continuous monitoring of human skin hydration and body sweat loss(2023)

Research Paper-3 Recent Advancements in Wearable Hydration-Monitoring Technologies: Scoping Review of Sensors, Trends, and Future Directions(2025)

Research Paper-4 Wearable Hydration Monitoring Technologies & AI-Based Hydration Intelligence(2024)

Research Paper 5: Review of Advances in the Measurement of Skin Hydration Based on Sensing of Optical and Electrical Tissue Properties(2022)

Research Paper 6: Characterizing Variability in Non-InvasiveHydration Monitoring Using Raman Spectroscopy(2025)

Research Paper 7:Multi-Modal Spectroscopic Assessment of Skin Hydration(2024)

Research Paper 8: An accurate wearable hydration sensor: Real-world evaluation of practical use(2022)

Research Paper 9: Noninvasive Monitoring to Detect Dehydration: Are We There Yet?(2023)

Research Paper 10: Hydration Assessment Using the Bio-Impedance Analysis Method(2022)

From the references of these Research papers:

Instead of giving everyone the same generic "drink 3 liters of water a day" rule, Get Vari looks at hydration dynamically. How much water you need changes by the minute based on your body, your work, and the weather. To calculate this accurately, our system splits data into two categories: **Primary Parameters** (live sensor data) and **Secondary Parameters** (your personal profile - combination of physical and physiological inputs).

1.1 Primary Parameters

These numbers are tracked in real-time by the sensors built into the Get Vari wristband, powered by an **ESP32 Microcontroller**:

1. **Heart Load (HL):** Tracked by the **Pulse Sensor MAX86141**. When your body runs low on water, your blood gets slightly thicker, and your heart has to pump faster to keep up. A rising heart rate is our clearest internal warning sign.
2. **Activity Load (AL):** Tracked by the **MPU6500 Motion Sensor** (Accelerometer and Gyroscope). It detects whether you are sleeping, sitting, walking, or working out so it can estimate how much you are sweating.
3. **Temperature Load (TL):** Tracked by the **MAX30208 Sensor**. It measures how hot the air is around you, which tells the system if your body is actively sweating to cool down.
4. **Humidity Load (HuL):** Tracked by the **AHT10 Sensor**. High humidity stops sweat from evaporating off your skin. This creates "silent dehydration" because you keep losing water even if you don't feel dry.
5. **Time Load (TiL):** A built-in timer that counts how many hours have passed since you last logged drinking water. It keeps you safe even when you are resting in a comfortable room.

ESP32 calculates:

$$\text{HRisk} = f(\text{HL}, \text{AL}, \text{TL}, \text{HuL}, \text{TiL})$$

Where:

- HL = Heart Load
- AL = Activity Load
- TL = Temperature Load
- HuL = Humidity Load
- TiL = Time Load

1.2 Secondary Parameters

Research paper: Water, Hydration and Health(2010, Aug)

Research Paper: A review of permissible limits of drinking water(2012)

Research Paper: Dietary Reference Intakes for Water, Potassium, Sodium, Chloride, and Sulfate (2005)

Research Paper: The water use of Indian diets and socio-demographic factors related to dietary blue water footprint(2017)

According to these research papers:

1. One should drink 7-8 glasses of water everyday i.e. 2-3 L of water in a day.
2. The Adequate Intake (AI) for total water intake for young men and women (ages 19 to 30 years) is 3.7 L and 2.7 L per day, respectively.
3. Fluids (drinking water and beverages) provided 3.0 L (101 fluid oz; $\approx$ 13 cups).
4. 2 L (74 fluid oz; $\approx$ 9 cups) per day for 19- to 30-year-old men and women, respectively, representing approximately 81 percent of total water intake in the U.S. survey.
5. The primary indicator of hydration status is plasma or serum osmolality.
6. Because normal hydration can be maintained over a wide range of water intakes, the AI for total water (from a combination of drinking water, beverages, and food) is set based on the median total water intake
7. Silent Dehydration” is a critical phenomenon in coastal cities (Mumbai, Chennai, Kochi) where high ambient humidity prevents sweat from evaporating.
8. Cold Areas like Leh, Himachal, people drink less water because they don't feel thirsty.

Factor	Impact on Water Need
Climate	Hotter regions require more water
Humidity	High sweating → more water loss
Physical activity	Exercise increases sweat loss
Age	Elderly have low thirst sensation
Gender	Men generally need more water
Diet	High salt/protein diets need more water
Disease conditions	Kidney, diabetes, BP patients need monitoring
Altitude	High altitude causes faster water loss
Occupation	Outdoor workers need more hydration

Important Insights for Different Age Groups

Group	Main Risk	Key Insight
Children	Heat Stress	Don't recognize thirst
Teenagers	Sports Dehydration	Performance drops
Adults	Work stress + low intake	Chronic dehydration
Elderly	Reduced thirst sensation	Silent dehydration

Main Factors Affecting Water Consumption

Factor	Impact on Water Consumption
Region	Strongest factor affecting water usage
Diet type	Rice/wheat-heavy diets increase WF
Urbanization	Urban diets use more water
Income-Level	Higher income → higher water footprint
Gender	Males consume more water-intensive diets
Age	Younger people had higher dietary WF
Religion	Non-Hindu diets had higher WF due to more animal food consumption
Education	Higher education linked to higher WF
Meat Consumption	Increases water usage

Water Footprint (WF) of Indian Diets

The paper calculates how much water is indirectly consumed through food.

Type of water footprint	Average consumption
Blue WF (Irrigation Water)	737 liters/person/day
Green WF (Rainwater)	2531 liters/person/day
Total WF	~3268 liters/person/day

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Regional Insights

Region	Water Consumption Insight	Main Reason
North India	Highest blue WF	Heavy wheat consumption
South India	Lowest blue WF	Higher rice consumption
West India	Very high blue WF	Wheat + oil-heavy diets
East India	Moderate WF	Mixed diet patterns

Note: North India uses ~1.5x more Blue Water(Irrigation Water)

Urban and Rural Water Consumption

Category	Water Consumption
Urban Population	High Water
Rural population	Low Water

Urban populations:

- 1. Consume more processed foods
- 2. Consume more dairy/poultry
- 3. Have higher calorie intake
- 4. Have higher living standards

Secondary Parameters for Get Vari Hydration Intelligence System

Secondary parameters are user-specific characteristics and lifestyle factors that indirectly influence hydration status. These parameters help the Get Vari algorithm create a personalized hydration profile and improve the accuracy of hydration-risk prediction.

Secondary Parameter	Why It Is Important
Age	Different age groups have different water requirements and dehydration susceptibility.
Gender	Males and females have different body water percentages and fluid regulation mechanisms.
Height	Height contributes to body composition calculations and hydration estimation.
Weight	Body weight influences daily water requirements and sweat-loss calculations.
Occupation	Different occupations expose users to varying levels of heat, activity, and dehydration risk.
Daily Activity Level	Physically active individuals lose more fluids through sweating and require increased hydration.
Exercise Frequency	Frequent exercise increases daily water loss and hydration requirements.

Exercise Intensity	Higher exercise intensity causes greater sweat production and electrolyte loss.
Outdoor Exposure Duration	Longer exposure to sunlight and heat increases dehydration risk.
Geographic Location	Climate and environmental conditions influence water loss and hydration needs.
Diabetes	Diabetes can increase urination frequency, leading to greater fluid loss.
Kidney Stones	Individuals with kidney stones often require higher fluid intake to reduce recurrence risk.
Chronic Kidney Disease (CKD)	Kidney disorders affect fluid regulation and require personalized hydration recommendations.
Heart Disease	Cardiovascular conditions influence fluid balance and hydration management.
High Blood Pressure	Blood pressure can be affected by hydration status and fluid intake.
Thyroid Disorders	Thyroid conditions can affect metabolism, sweating, and body temperature regulation.
Digestive Disorders	Vomiting and diarrhea can rapidly cause dehydration.
Daily Water Intake	Current water-consumption habits help establish hydration baselines.
Water Reminder Dependency	Users who often forget to drink water may require more proactive alerts
Water Bottle Carrying Habit	Indicates the user's hydration awareness and behavioral patterns.
Workplace Environment	Indoor, outdoor, or high-temperature workplaces affect dehydration risk differently.
Sweating Pattern	Heavy sweaters lose fluids and electrolytes more rapidly than others.
Morning Thirst	Waking up thirsty can indicate overnight dehydration.
Caffeine Consumption	Excessive caffeine intake may contribute to increased fluid loss.
Energy Drink Consumption	Energy drinks can influence hydration and cardiovascular responses.

Alcohol Consumption	Alcohol promotes fluid loss and increases dehydration risk.
Dietary Pattern	Food choices affect electrolyte balance and hydration requirements.

Role of Secondary Parameters

Unlike primary parameters, which are measured continuously through sensors, secondary parameters remain relatively stable and are collected during user registration. These factors act as personalization modifiers within the Get Vari Hydration Intelligence Algorithm and allow the system to generate user-specific hydration-risk scores, water-intake recommendations, LED alerts, vibration notifications, and mobile-app guidance.

Section 2: The Core Formulas & Everyday Use Cases

2.1 The General Base Formula

For a healthy adult with no medical issues, the system calculates your **Hydration Risk Score** on a scale from 0 (Fully Hydrated) to 100 (Critical Risk) using this balanced baseline equation:

$$\text{Hydration risk} = 0.30(\text{Heart Load}) + 0.25(\text{Activity Load}) + 0.20(\text{Temp. Load}) + 0.10(\text{Humidity}) + 0.15(\text{Time Load})$$

Cohort 1: Corporate Office Worker (Air-Conditioned Office)

- **Why the formula changes:** Air conditioning chills the skin and hides sweat, tricking your brain into thinking you aren't thirsty. High mental workload also makes you forget to drink. Therefore, the timer (TiL) is given half the total weight (50%).
- **The Model:**
$$\text{HRiskOffice} = 0.15(\text{HL}) + 0.10(\text{AL}) + 0.15(\text{TL}) + 0.10(\text{HuL}) + 0.50(\text{TiL})$$
- **Real-Life Scenario:** You sit at a desk in a cool 21°C office in Bengaluru for 6 hours straight. Your heart rate is low, and you aren't moving. But because Get Vāri weights the timer at 50%, your risk score will slowly climb, sending you a friendly reminder to drink before you get a dehydration headache.

Cohort 2: Outdoor / Field Worker

- **Why the formula changes:** If you work out in the sun, heat (TL) and physical movement (AL) are your biggest threats. Time doesn't matter here—even if you drank water 15 minutes ago, walking in the sun means you are losing fluid rapidly.
- **The Model:**
$$\text{HRiskOutdoor} = 0.20(\text{HL}) + 0.35(\text{AL}) + 0.35(\text{TL}) + 0.05(\text{HuL}) + 0.05(\text{TiL})$$
- **Real-Life Scenario:** A field manager is walking a construction site in Rajasthan at 42°C. The weather and motion parameters (weighted heavily at 35% each) instantly drive the risk score up, alerting them to drink large amounts immediately to prevent sudden heat exhaustion.

Cohort 3: Diabetic Patient

- **Why the formula changes:** High blood sugar forces the kidneys to dump water constantly. Missing a water window can quickly cause your blood to thicken, making blood sugar levels spike even higher.
- **The Model:**
$$\text{HRiskDiabetic} = 0.25(\text{HL}) + 0.20(\text{AL}) + 0.15(\text{TL}) + 0.10(\text{HuL}) + 0.30(\text{TiL})$$
- **Real-Life Scenario:** A diabetic user goes a few hours without water. The elevated timer weight (30%) and heart weight (25%) work together to trigger an early alert, prompting them to sip water frequently to keep blood sugar safely diluted.

Cohort 4: Chronic Kidney Disease (CKD) Patient

- **Why the formula changes:** Weak kidneys cannot process or get rid of excess water. Over-drinking can put dangerous fluid pressure on the lungs and heart. The formula flips its priorities: it lowers the timer weight so the app doesn't push them to drink, and raises the heart weight (35%) to monitor for internal fluid retention and heart strain.
- **The Model:**
$$\text{HRiskKidney} = 0.35(\text{HL}) + 0.15(\text{AL}) + 0.25(\text{TL}) + 0.15(\text{HuL}) + 0.10(\text{TiL})$$
- **Real-Life Scenario:** A kidney patient has already had their limited daily allotment of water. If the band detects their heart rate rising while they are sitting completely still, it recognizes fluid buildup strain and warns them to rest and stop drinking.

Cohort 5: Kidney Stone Patient

- **Why the formula changes:** To stop stones from forming, the kidneys must constantly be flushed with water. The timer weight is raised to 30% to make sure the user drinks at fixed intervals throughout the day.
- **The Model:**
$$\text{HRiskStone} = 0.20(\text{HL}) + 0.20(\text{AL}) + 0.20(\text{TL}) + 0.10(\text{HuL}) + 0.30(\text{TiL})$$

Section 3: The Hydration Risk Algorithm

Normalize Sensor inputs:100-115

1. Heart Load

Heart Rate	Physiological state	Heart Load
<55 BPM	Deep recovery/rest	5
55-70 BPM	Normal resting	15
70-85BPM	Mil activation	30
85-100 BPM	Elevated strain	50
100-115 BPM	High cardiovascular load	70
115-130	Severe exertion	85
>130	Critical stress	100

2. Activity Load

Activity State	Motion Interpretation	Activity load
Sleeping	No-Movement	0-5
Sedentary	Desk Work	10
Light walking	Daily Movement	25
Moderate movement	Active routine	45
Workout	Gym exercise	65
High cardio	Running/ Training	85
Extreme exertion	Sprint/ heat training	100

3. Temperature Load:

Ambient Temp	Thermal stress	Temp Load
<18°C	Cool	5
<18-24°C	Comfortable	15
<24-28°C	Warm	35
<28-32°C	High heat	65
<32-36°C	Severe Heat strain	80
>36°C	Critical thermal load	100

4. Humidity Load

Humidity	Physiological Effect	Humidity Load
<30%	Dry Air	10
30-45%	Comfortable	20
45-60%	Mild Sweat	35
60-75%	Impaired cooling	55
75-85%	Severe evaporation reduction	75
>85%	Dangerous Humidity	95

5. Time Load:

Time since water intake	Physiological risk	Time load
<20mins	Fully hydrated	0
20-60 mins	stable	10
1-2 hrs	Mild deficit	25
2-4 hrs	Moderate deficit	45
4-6 hrs	High dehydration policy	65
6-8 hrs	Severely dehydration risk	80
>8 hrs	Critical hydration neglect	100

Hydration Status:

Score	Status
0-25	Hydrated
26-50	Mild Risk
51-75	High Risk
76-100	Critical

Water Intake Recovery Table

Water Intake	Base Reduction	Actual reduction during high stress
100ml	3-5	2-4
250ml	8-12	6-9
500ml	15-20	12-15
750ml+	20-30	15-22

Absorption takes:

1. 10-45 mins of absorbing water
2. Depends on activity
3. Depends on stomach emptying
4. Depends on sweating rate

The complete timeline for this multi-compartment stabilization typically spans **45 to 60 minutes**. This biological delay is directly validated by clinical literature. **According to a comprehensive research paper titled "*Fate of ingested fluids: factors affecting gastric emptying and intestinal absorption of beverages in humans*" authored by John B. Leiper (2015) in the journal *Nutrition Reviews***, water is initially stored in the stomach with little to no net absorption occurring across the gastric mucosa. The study establishes that the total time required for a bolus of water to empty into the small intestine and fully enter systemic blood circulation sits strictly within a **10-to-45-minute window**, a variable that is heavily governed by the total volume and formulation of the liquid consumed.

Explanation for Water Intake Recovery:

1. Base Reduction (Resting / Low Stress State)

This represents the points deducted from the Risk Score when the body is operating under normal, stable conditions (e.g., sitting in a temperature-controlled office, normal resting heart rate).

The Physics: Because the body is at rest, the gastrointestinal (GI) tract receives maximum blood flow, allowing for optimal gastric emptying and rapid intestinal absorption. The body gets the full rehydration benefit of the water.

2. Actual Reduction During High Stress (Active / Strain State)

- a. This represents the points deducted when the Get Vari sensors detect elevated Heart Load (HL), high Activity Load (AL), or severe Temperature Load (TL).
- b. The Physiology: Under physical or thermal stress, the sympathetic nervous system triggers a fight-or-flight response, shunting blood flow away from the stomach and digestive organs and directing it toward working muscles and skin for sweating.
- c. The Consequence: Gastric emptying slows down drastically. Even though the water is in the stomach, it is not yet in the bloodstream. Furthermore, the user is likely continuing to lose water through sweat *while* drinking. Therefore, the system applies a penalized, conservative reduction value

Scenario A: The Sedentary Desk Worker (Base Reduction in Action)

1. **User Profile:** An IT professional sitting in a cool, air-conditioned office in Bengaluru.
2. **Current State:** They haven't drank water in 4 hours. Their Risk Score has steadily climbed to **45 (Mild Risk)** primarily driven by Time Load (TiL).
3. **Action:** They drink a standard **250ml** glass of water.
4. **System Logic:** Get Vari detects a resting heart rate (65 BPM) and a comfortable temperature (22°C). The algorithm pulls from the **Base Reduction** column and subtracts the maximum **12 points**.
5. **Result:** Their risk drops from **45 down to 33 in 1 hour**, safely returning them toward the fully hydrated zone.

Scenario B: The Sweltering Delivery Executive (High-Stress Reduction in Action)

1. **User Profile:** A logistics delivery partner riding a motorbike in Delhi during peak summer heat (40°C).
2. **Current State:** Their Risk Score is sitting at a dangerous **72 (High Risk)** due to massive Temperature Load (TL) and an elevated Heart Load (HL) of 110 BPM.
3. **Action:** They stop at a roadside stall and drink an identical **250ml** glass of water.
4. **System Logic:** Get Vari's multi-sensor fusion sees the 40°C ambient heat and the 110 BPM heart strain. It knows the user's body is actively fighting to cool itself down and that intestinal absorption is severely impaired. Instead of a 12-point reduction, it applies the conservative **High Stress Reduction of 6 points**.
5. **Result:** Their risk score drops from **72 down to 66**. It intentionally leaves them in the high-risk zone because a mere 250ml is not enough to counteract the ongoing, massive sweat rate and internal cardiovascular drift they are experiencing. The wearable will quietly nudge them to drink another 250ml within the next few minutes.

Section 4: Profile Details: What We Collect & Why

4.1 Personal & Lifestyle Basics

- **Full Name:** Used to set up your personal account and greet you within the app.
- **Age:** Different age groups handle water differently. For example, children lose fluids very quickly, while older adults often have a weaker natural sense of thirst.
- **Gender:** Males generally have a higher percentage of lean muscle, which naturally holds more body water, while females generally have a lower body water percentage. The app uses gender-specific baselines to stay accurate.
- **Height and Weight:** Larger individuals or those with more body mass require more water daily to stay cool and keep their metabolism running smoothly.
- **City and Location:** The app uses your location to check local weather data. Living in a hot or highly humid region completely changes how fast your body loses water.
- **Occupation & Daily Activity Level:** Your daily job determines your physical effort. An office worker sitting in a chair has a much lower dehydration risk than a construction worker or field engineer working out in the sun.
- **Outdoor Exposure & Exercise Habits:** The app asks how long you spend in direct sunlight and how often/intensely you work out. High-intensity exercise in the sun causes rapid fluid loss through sweat, which requires faster tracking.

4.2 Medical & Health Background

This is the most critical part of the registration. Certain medical conditions directly change how your body processes fluids:

- **Diabetes:** High blood sugar forces the kidneys to flush out water much more frequently, causing rapid dehydration.
- **Kidney Stones:** People prone to kidney stones must maintain high fluid intake to flush their kidneys and prevent stones from forming.
- **Chronic Kidney Disease (CKD):** Weak kidneys cannot handle heavy amounts of water. For these users, the app adjusts carefully to ensure they do not accidentally over-hydrate, which can cause dangerous fluid buildup.
- **Heart Disease & Blood Pressure Conditions:** Fluid volume in the blood directly affects your blood pressure and heart strain. Keeping a steady fluid balance protects your cardiovascular system.
- **Medication Usage:** Common medications like diuretics (water pills), blood pressure drugs, and certain diabetes treatments actively force your body to lose more water through urination.

4.3 Symptom & Hydration Behavior Assessment

- **Current Symptoms:** The app checks if you frequently suffer from headaches, dizziness, fatigue, dry mouth, muscle cramps, or dark urine. These clues help the system recognize if you are already living in a state of chronic dehydration.
- **Hydration Habits:** We ask how much water you currently drink, if you use reminders, how often you feel thirsty, and whether you carry a water bottle. This helps the AI understand your behavioral baseline.
- **Workplace Climate:** Users select their typical workspace, whether it is an air-conditioned office, a non-AC indoor area, a shaded outdoor zone, or direct sunlight. Since air conditioning suppresses your natural thirst, this selection changes how the app times your notifications.
- **Sweating Patterns:** People who naturally sweat heavily lose both water and vital electrolytes at an accelerated rate, requiring earlier interventions.
- **Dehydration History:** If you have a history of severe dehydration, heat exhaustion, or heat stroke, the algorithm marks you as a higher-risk user for safety.

4.4 Personalization & Everyday Habits

- 1) **Primary Health Goal:** You can tell the app if your main goal is to stay generally hydrated, improve your athletic fitness, prevent heat stress at work, or manage a specific medical condition.
- 2) **Notification Preferences:** For alert messages, LED lights on the band (Red and Blue), physical wrist vibrations to drink water (5 sec of vibrations), smart mobile notifications.
 - a. **Red Light:** Highly Dehydrated (When Hydration Risk is critical)
 - b. **Blue Light:** Normal (When Hydration risk is normal)

4.5 Wristband Alerts & Water Absorption Dynamics

When the formula combines these points and calculates a final risk score, the wristband and mobile app change their behavior to give you clear, tiered warnings.

Table 4.1: Wristband Alert Levels & Responses

Your Risk Score	What It Means	OLED Screen Light	Wristband Vibration	What the App Tells You
0 to 25	Hydrated	Blue Colour	Completely Silent	Everything looks great! Your body is stable.
26 to 50	Mild Risk PDF	—	One Quick Buzz (1s)	Take a small sip (100ml–250ml)
51 to 75	High Risk	—	Pulsing Vibration (3s)	High Risk: Drink a full glass of water (250ml–500ml).
76 to 100	Critical Risk	Red Colour	Continuous Vibration (6s)	Danger: Drink 500ml slowly. Stop running/working

The Safe State (Score 0 – 25)

- **System Hydration Status: Hydrated**
- **Why the math puts a user here:** This score means your primary sensor parameters are extremely low. The user has likely drank water recently (low Time Load), is resting comfortably in a cool room (low Temp and Activity Loads), and their heart rate is perfectly normal (low Heart Load).
- **Hardware Behavior:** The OLED display shows a quiet, reassuring **Solid Green Ring**. The wristband motor stays **Completely Silent**.
- **App Behavior:** Displays a simple "System Nominal" dashboard. It doesn't disturb the user because no behavioral intervention is required.

The Attention State (Score 26 – 50)

- **System Hydration Status: Mild Risk**
- **Why the math puts a user here:** The score begins to tick upward. This typically happens when an IT worker has been sitting at a desk for 3–4 hours without drinking (Time Load rises to 45), or someone is taking a light stroll outside on a warm day.
- **App Behavior:** The app suggests a small, preventative volume: **"Sip 100ml – 250ml"** (a few swallows or a single glass). This stops dehydration early before any physical performance drops occur.

The Active Action State (Score 51 – 75)

- **System Hydration Status: High Risk**
- **Why the math puts a user here:** This is a serious pivot point. To cross a score of 50, multiple high-impact parameters must fire together. For example, a user is actively working out (Activity Load = 65), causing their heart rate to enter an active cardiovascular drift zone of 105 BPM (Heart Load = 70).
- **App Behavior:** Sends a high-priority **"High Alert"** push notification telling the user to drink a significant amount (**250ml – 500ml**). The app knows the sweat rate is outpacing current fluid absorption, requiring immediate intervention.

The Emergency State (Score 76 – 100)

- **System Hydration Status: Critical Risk**
- **Why the math puts a user here:** The user's body is experiencing extreme physical and biological crisis. They are either pushing maximum exertion in heavy fitness training (AL>85,HL>85) or are exposed to highly dangerous outdoor heat waves (TL=100). Blood plasma volume has dropped, causing the heart to pump at dangerously high rates to sustain cardiac output.
- **Hardware Behavior:** The OLED shows a bright **Solid Red Cross**. The haptic motor activates an intense, **Continuous 6-second Vibration**.
- **App Behavior:** Activates an **"Emergency Override."** It strictly commands an immediate **500ml intake** and explicitly prints: *"Advise slowing down physical activity."* It steps away from just recommending water and begins managing user safety to prevent heat stroke or severe muscle cramping.

Section 5: CASE STUDIES

This overview matches our core user personas to their exact formula models and target liquid goals.

Table 5.1: Master User Profiles & System Behavior

Who is the User	Main Sensor Looked At	Personalized Formula Applied	Why This Profile Matters	Target Daily Fluid
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IT Desk Professional	Timer (TiL)	$0.15(\text{HL})+0.10(\text{AL})+0.15(\text{TL})+0.10(\text{HuL})+0.50(\text{TiL})$	AC rooms hide sweat and blunt your natural thirst.	1.5–2.5 L/day
Field Logistics Agent	Weather (TL) & Motion (AL)	$0.20(\text{HL})+0.35(\text{AL})+0.35(\text{TL})+0.05(\text{HuL})+0.05(\text{TiL})$	High heat and walking cause rapid, massive fluid drainage.	3.5–6.0 L/day
Diabetic Patient	Timer (TiL)	$0.25(\text{HL})+0.20(\text{AL})+0.15(\text{TL})+0.10(\text{HuL})+0.30(\text{TiL})$	Protects the user from frequent, dehydration-driven urination cycles.	3.0–5.0 L/day
Kidney Failure (CKD)	Heart Rate (HL)	$0.35(\text{HL})+0.15(\text{AL})+0.25(\text{TL})+0.15(\text{HuL})+0.10(\text{TiL})$	Tracks heart signals to guard against dangerous fluid overload.	Set strictly by their doctor
Kidney Stone Patient	Timer (TiL)	$0.20(\text{HL})+0.20(\text{AL})+0.20(\text{TL})+0.10(\text{HuL})+0.30(\text{TiL})$	Forces steady water intake to stop minerals from forming into stones	High-volume flushing targets
Pregnant User	Heart Rate (HL)	$0.35(\text{HL})+0.15(\text{AL})+0.20(\text{TL})+0.10(\text{HuL})+0.20(\text{TiL})$	Tracks early heart drift to ensure stable blood flow for the baby.	Increased baseline fluids required

Section 6: FUTURE SCOPE

Future Scope and Product Roadmap

Characterizing Variability in Non-Invasive Hydration Monitoring Using Raman Spectroscopy" (Rourke-Funderburg et al., 2025)

This experimental study explores the use of **Raman spectroscopy as a non-invasive technique for monitoring human hydration status by analyzing molecular changes in biological tissues**. The research investigates how Raman spectral patterns vary with different hydration levels and evaluates the potential of spectroscopy-based approaches for detecting hydration-related changes without

requiring blood or urine samples. The study highlights that Raman spectroscopy can identify biochemical variations associated with tissue water content; however, individual variability, skin properties, physiological differences, and environmental factors create challenges in achieving accurate hydration estimation. The findings suggest that Raman-based hydration monitoring requires **advanced calibration methods, machine learning/AI algorithms, and personalized models** to improve reliability for real-world wearable applications.

This research supports the idea that hydration cannot be estimated using a single fixed water-intake recommendation; instead, dynamic hydration prediction requires combining multiple physiological and environmental parameters with AI-based interpretation.

To build upon the existing multi-sensor framework of the Get Vari wearable system, the future product roadmap focuses on integrating advanced non-invasive biochemical sensors, hardware expansions, and deeply personalized artificial intelligence engines. By shifting from localized physiological loads to direct cellular and fluid compartment metrics, upcoming iterations of the platform will implement the following key enhancements:

1. **Advanced Sweat Analysis and Electrolyte Monitoring:** Future versions of the wearable hardware will incorporate specialized electrochemical sweat sensors to continuously measure vital electrolyte concentrations, including dedicated sodium, potassium, and chloride tracking. This will allow the system to detect real-time salt depletion alongside pure water loss, helping athletes and outdoor workers mitigate the risk of muscle cramps and electrolyte imbalances.
2. **Bioimpedance Spectroscopy Integration:** The device architecture will be expanded to include Bioelectrical Impedance Analysis (BIA) and capacitance-based tissue sensing. By passing an imperceptible, low-frequency electrical signal directly through the wrist tissue, the system will be able to map changes in skin resistance and capacitance to calculate Total Body Water (TBW), providing a direct physiological validation of intracellular and extracellular fluid shifts.
3. **AI-Based Hydration Prediction Engine:** Moving beyond real-time risk calculations, the cloud infrastructure will deploy advanced predictive machine learning models, such as Random Forest and deep learning architectures, to forecast dehydration events up to an hour before they manifest physically. This personalized hydration intelligence engine will leverage historical behavioral data, baseline user demographics, and personal health habits to dynamically adjust risk coefficients over time.
4. **Smart Water Bottle and Weather-Aware Integration:** To close the loop on logging accuracy, the system will feature native integration with smart ecosystem

hardware, such as IoT-enabled water bottles, to automatically capture and verify exact water intake volume without requiring manual app entries. Furthermore, by deeply incorporating predictive, localized weather API metrics, the platform will deliver context-aware, forward-looking hydration schedules tailored to upcoming regional micro-climate shifts and extreme thermal exposures.